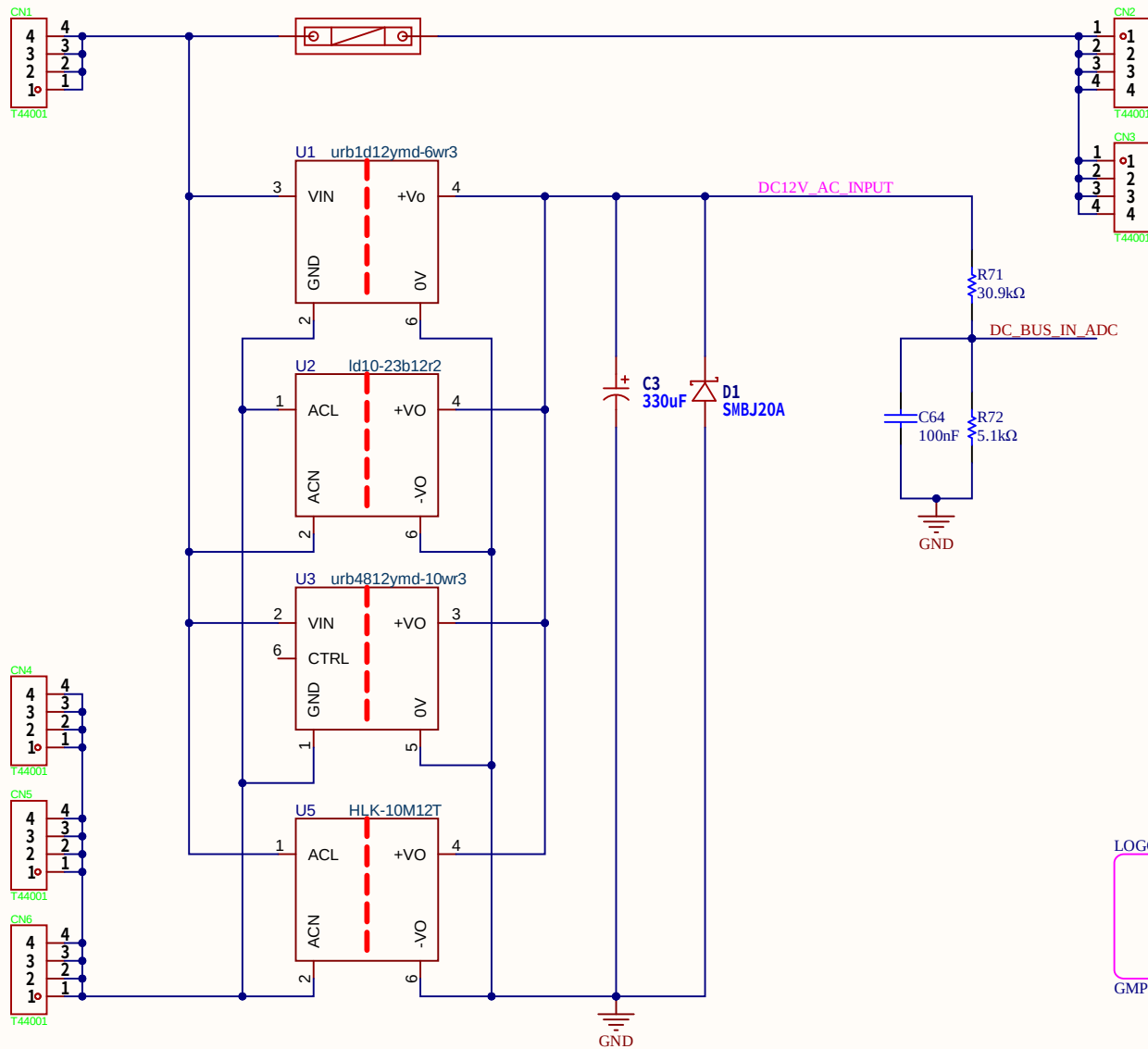




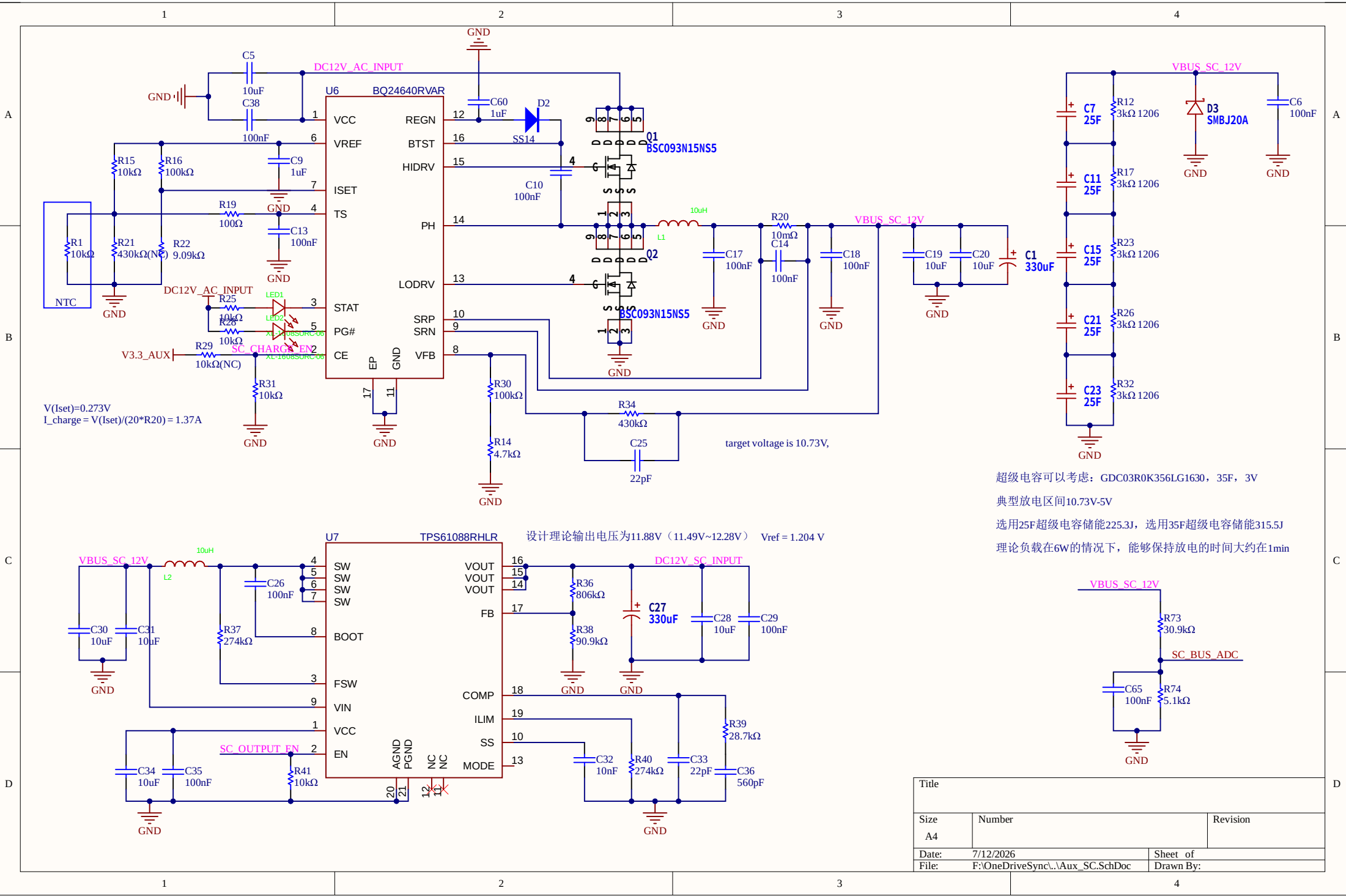
LOGO1
GMP_LOGO
GMP LOGO

LOGO3
JsScript_LOGO
JsScript_LOGO

LOGO2

DANGER HIGH VOLTAGE

Title		
Size A4	Number	Revision
Date: 7/12/2026	Sheet of	
File: F:\OneDriveSync\Main_AC.SchDoc	Drawn By:	



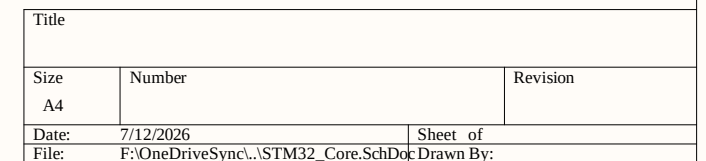
$$V(I_{set}) = 0.273V$$
$$I_{charge} = V(I_{set}) / (20 * R20) = 1.37A$$

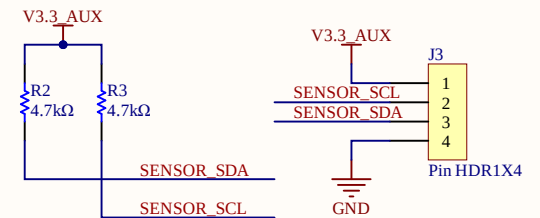
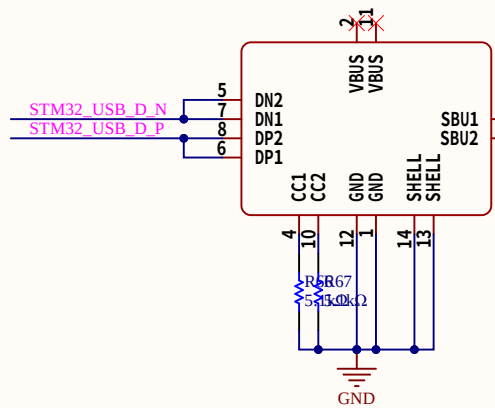
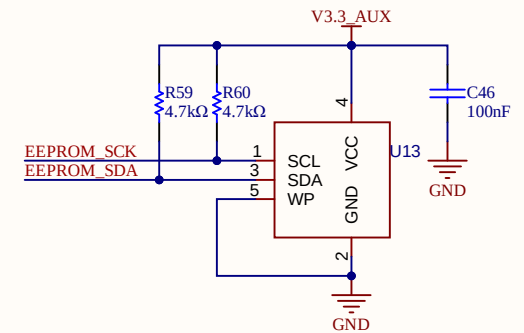
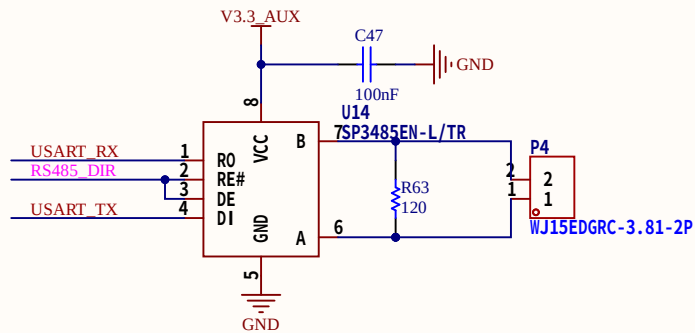
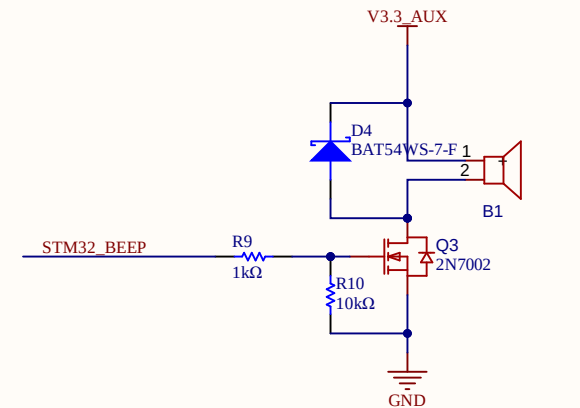
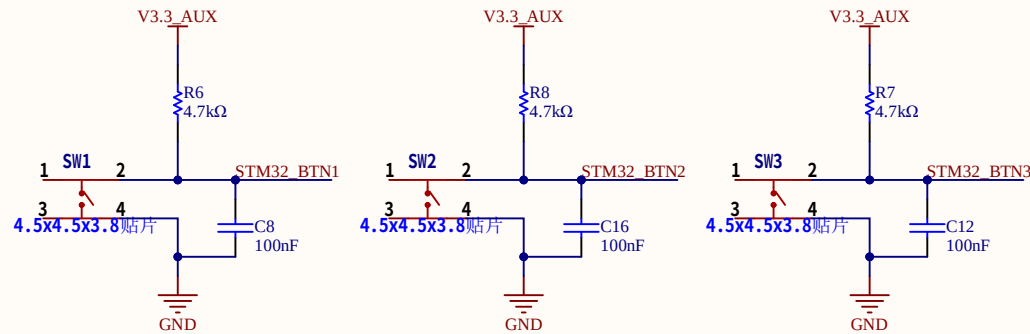
设计理论输出电压为11.88V (11.49V~12.28V) Vref = 1.204 V

target voltage is 10.73V,

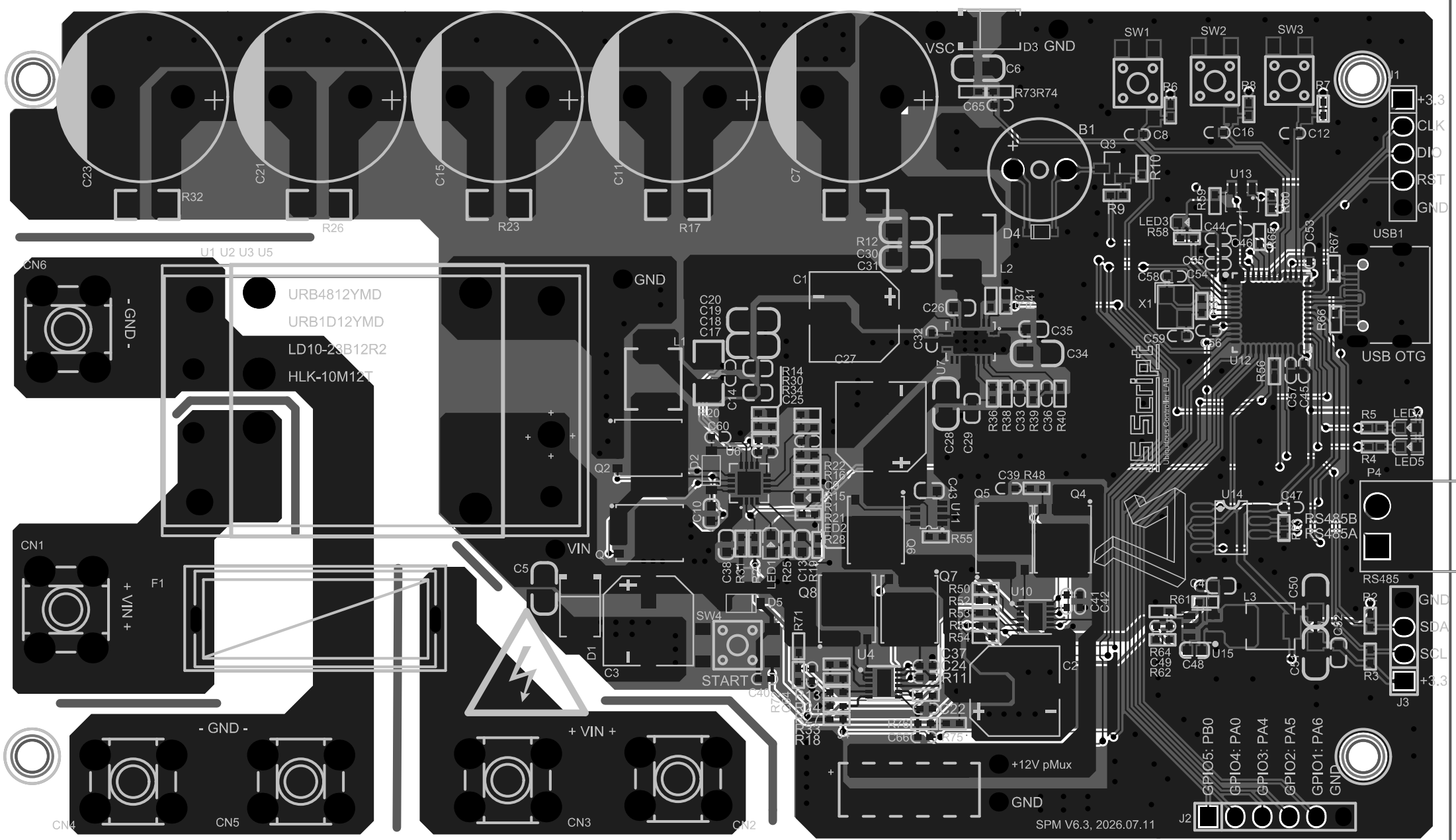
超级电容可以考虑: GDC03R0K356LG1630, 35F, 3V
典型放电区间10.73V-5V
选用25F超级电容储能225.3J, 选用35F超级电容储能315.5J
理论负载在6W的情况下, 能够保持放电的时间大约在1min

Title		
Size	Number	Revision
A4		
Date:	7/12/2026	Sheet of
File:	F:\OneDriveSync\...\Aux_SC.SchDoc	Drawn By:





Title		
Size	Number	Revision
A4		
Date:	7/12/2026	Sheet of
File:	F:\OneDriveSync\STM32_Peripheral_Schematic	Drawn By:



Comment	Description	Designator	Footprint	LibRef	Quantity
#H8080A05	buzzer	B1	BLZ-TH-8080-A- P360-101-Y-10	#H8080A05	1
330uF		C1, C3, C38, C37	CAP-350V-ROB-D- 8.3-V08.3-1F	PV1E1331M0810 330uF-25V	4
100nF	无极性电容	C4, C10, C13, C17, C18, C28, C29, C35, C38, C43, C48, C57	D603C	CAP	12
10uF	无极性电容	C5, C19, C20, C28, C30, C31, C34, C36, C31	D855C	CAP	9
100nF	无极性电容	C7, C11, C15, C21, C29	D855C	CAP	1
22uF		C8, C12, C14, C16, C24, C37, C40, C41, C42, C44, C46, C47, C53, C54, C56, C56, P48, C58, C56	CAP-TH-ROB-D- P750-D1.2-1F	BPF1VM222L250T	5
100nF	无极性电容	C2, C3, C37, C40, C41, C42, C44, C46, C47, C53, C54, C56, C56, P48, C58, C56	D402C	CAP	19
1uF	无极性电容	C9, C39	D402C	CAP	2
100nF	无极性电容	C22, C39, C39	D402C	CAP	3
22uF	无极性电容	C25, C33, C49, C49, C59	D402C	CAP	5
100nF	无极性电容	C38	D402C	CAP	1
4.7uF	无极性电容	C45, C57	D402C	CAP	2
T44001		CN1, CN2, CN3, CN4, CN5, CN6	CONN-TH-144001	T44001(PCB插子)	6
SMB020A	C151257 (NoValue)	D1, D3	SMB-1.6-WS6- 8.5-A-10	SMB020A_C151257	2
0E14	普通二极管	D2, D5	DOU-133	DIODE	2
BA7S48W5-7-F	晶体管—偏置	D4	SCD-323	DIODE-SCHOTTKY	1
模拟滤波用途	模拟滤波用途	F1	FUSE-TH-1245- W9.5-P22.5	模拟滤波用途	1
Pin-HER13X5		J1	Pin-HER13X5	Pin-HER13X5	1
Pin-HER13X6		J2	Pin-HER13X6/2.54mm-S	Pin-HER13X6	1
Pin-HER13X4		J3	Pin-HER13X4/2.54mm-S	Pin-HER13X4	1
PZ254-2-05-5	HER-2.54-2*5P	J4	HER-2.54-2*5P PZ254-V-M-R2-C5- 1.7A	PZ254-2-05-5	1
10uH		L1, L2	IND-SMD-15.7- W5.2-5MM1020	IND0600-3R3-M	2
3.3uH		L3	IND-SMD-14.4- W5.2	IND0600-3R3-M	1
XL-16080URC-06	LED D603	LED1, LED2, LED3, LED4, LED5	LED0603-RO-RED	XL-16080URC-06	5
uMP-LOGO	LOGO NAME Main	LOGM1	uMP-LOGO	uMP-LOGO	1
u30mp1_LOGO	LOGO 300mp1	LOGO3	u30mp1-R-V2-1001- 0508	u30mp1_LOGO	1
W1516DCRC-3.81- 2P	C3837	P4	CONN-TH-2P- P3.81-W1516DCRC- 1.81-2P	W1516DCRC-3.81- 2P	1
80C08JN10N15	MCSPET minicon- 130V, 17A, 9.3mC 80C08-45A	Q1, Q2, Q4, Q5, Q6, Q7, Q8	PQ-1050N-8-1.5-D- W6.0-PL27-136.2- 86-4P	80C08JN10N15	7
2N7002	MCSPET	Q3	SCOT-23-3-L2.9- W1.3-P1.50-132.4- 88	2N7002	1
10K()		R1, R10, R15, R18, R25, R28, R31, R41, R51, R55, R61, R64, R2, R5, R6, R7, R8, R14, R59, R59	D402R	RES	12
47K()		R14, R59, R59, R59	D402R	RES	8
10K()		R14, R59, R59, R59	D402R	RES	4
100K()		R11, R19, R48	D402R	RES	3
3K() 1206		R12, R17, R23, R25, R27	D402R	RES	5
100K()		R13, R38, R50	D402R	RES	3
100K()		R18, R19, R56, R65	D402R	RES	4
100uH()		R20	L206R	RES	1
450K(10K)		R31	D402R	RES	1
500K()		R22, R24, R52	D402R	RES	3
1.5K()		R27, R33	D402R	RES	2
10K(10K)		R29, R31, R54	D402R	RES	3
430K()		R34	D402R	RES	1
100K()		R36	D402R	RES	1
22K()		R37, R40	D402R	RES	2
10K(10K)		R39	D402R	RES	1
10K()		R67	D402R	RES	1
11K()		R62	D402R	RES	1
10K()		R63	D402R	RES	1
51K()		R66, R67, R72, R74, R75	D402R	RES	5
50K()		R71, R73, R75	D402R	RES	3
4.5uA5x3.89E片		SW1, SW2, SW3, SW4	SW-SMD-4P-1.45- 4.5A-5x3.89E片	4.5A5x3.89E片	4
ufb1d12ymd-6w3	最高值 6W DC-DC 转换器产品。输入 电压 40-160VDC。 额定输出 30W。输入 到输出效率测试电 压为 220VDC。可 在 -40°C 至 +85°C 的环境温度下安全 运行。具备输入欠 压保护、输出短路 过流、过压保护 功能。	U1	PWIR- TH-URB1DXXYMD- 6W3	ufb1d12ymd-6w3	1
W10-23B12()	最高值 输入电压范 围：交流两用。 额定值：恒流源模 式：高效率、高可 靠性、安全隔离、 广泛适用于工业 电力、发电、仪表 、通信及民用等行 业。	U2	PWIR-TH-LD1D- 23BX02	W10-23B12()	1
ufb4812ymd-10w3	最高值 输入电压范 围：交流两用。 额定值：恒流源模 式：高效率、高可 靠性、安全隔离、 广泛适用于工业 电力、发电、仪表 、通信及民用等行 业。	U3	PWIR- TH-URBXXYMD- 10W3	ufb4812ymd- 10w3, u5369780	1
LM74800QRRRQ1	Ideal Diode	U4, U10	HECON-12-L29- W3.0-P0.50-TL-EP- 2	LM74800QRRRQ1	2
AK-10M12T	最高值 输入电压范 围：交流两用。 额定值：恒流源模 式：高效率、高可 靠性、安全隔离、 广泛适用于工业 电力、发电、仪表 、通信及民用等行 业。	U5	PWIR-TH-4P- 230.15-W21.5-TH- LNK-4K-SMX08	W10-23B12()	1
BQ24640RVAR	PWR IC-Charger	U6	VQFN-16-L36- W3.6-P0.50-BL-EP	BQ24640RVAR	1
TP56108BPHLR	PWR Monitor 1.7V-12V -> 5.0V-12.6V, 80mA	U7	VQFN-20-L46- W3.6-P0.50-BL	TP56108BPHLR	1
LM74700QDPRQ1	Ideal Diode Driver	U11	TSOT-23-8-L2.9- W1.6-P0.65-L52.8- BL	LM74700QDPRQ1	1
STM32F411CEUE	STM32	U12	LFQFPN-48-L7.0- W7.0-P0.50-BL-EP	STM32F411CEUE	1
AT24C02D-STUM- T1-TUD1	Peripheral EEPROM 8kB	U13	SCOT-23-5-L3.0- W1.7-P0.95-L52.8- BL	AT24C02D-STUM- T1-TUD1	1
SP348SEN-LUTR		U14	SOIC-8-L5.0-W4.0- 2.2-TL-156.0-BL	SP348SEN-LUTR	1
TP5963201DOCR	PWR Buck 4.5-17V -> 0.768-7V, 3A	U15	SCOT-23-6-L2.9- W1.6-P0.95-L52.8- BL	TP5963201DOCR	1
TYPE-C 16PIN SMD0028	USB Type C 12P	U181	USB-C-SMD-TYPE- C-16PIN-SMD-12P- 1M600738	TYPE-C 16PIN	1
DCS_25M_responder	CISC passive 3225	X1	CRYSTAL-SMD-4P- 3.3-392.5-BL	DCS_25M_responder	1